

A. BABAEI

Neuroscience Researcher | Biomedical Engineer | Scientific Computing

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PROFESSIONAL PROFILE

PhD student in Neuroscience with an M.Sc. in Biomedical Engineering (Bioelectric) and multidisciplinary experience spanning experimental neuroscience, neural engineering, behavioral analysis, scientific programming, medical-equipment operations, technical education, and digital product platforms. Hands-on experience with rodent models, stereotaxic procedures, deep brain stimulation research, behavioral video analysis, MATLAB and Python workflows, machine learning, signal and image processing, and evidence synthesis. Experienced in converting complex scientific and technical work into reproducible analyses, publication-ready figures, structured courses, websites, and practical commercial outputs.

CORE EXPERTISE

- Experimental neuroscience and rodent disease models
- Stereotaxic procedures, DBS research, and behavioral testing
- Behavioral video scoring and event-based analysis
- Statistical modeling, reproducible workflows, and data visualization
- Python, MATLAB, machine learning, signal and image processing
- Scientific writing, evidence extraction, and graphical abstracts
- Medical and dental equipment inspection and troubleshooting
- Website administration, SEO, multimedia, and digital product operations

RESEARCH EXPERIENCE

Doctoral Research Project Current

Parafascicular Thalamic Deep Brain Stimulation in a Rodent Model of Parkinsonism

- Conducting experimental neuroscience research using a unilateral 6-OHDA rat model and parafascicular thalamic stimulation.
- Contributing to protocol design, lesion validation, electrode implantation workflows, behavioral testing, and scientific documentation.
- Developing behavioral data pipelines and repeated-measures statistical analyses in Python; preparing manuscript text, figures, and supplementary material while withholding unpublished quantitative results from public release.

EDUCATION

Iran University of Science and Technology 2023 - 2026

M.Sc. Biomedical Engineering (Bioelectric)

- Focus: neural engineering, machine learning, biomedical signal processing, and image processing.

Islamic Azad University, Tehran East Branch 2019 - 2022

B.Sc. Biomedical Engineering | GPA: 18.68/20

- Co-authored two ISC-indexed publications and contributed to launching the university scientific journal.

PROFESSIONAL EXPERIENCE

Peyramoon Teb Co. | Tehran Dec 2023 - Mar 2026

Medical Equipment Supervisor Engineer (Part-Time)

- Performed equipment supervision, routine inspections, compliance checks, technical assessment, and documentation.
- Supported installation, preventive maintenance, and troubleshooting in clinical settings.

Vistatech-Pishro & Vista Co. | Tehran Mar 2023 - May 2023

Medical Equipment Repair Technician (Internship to Specialist Assignment)

- Completed practical training in CSR and sterilization equipment, dental handpieces, autoclaves, ultrasonic cleaners, and basic laboratory devices.

Sirdent.ir Apr 2020 - Present	Founder and Website Administrator - Online Medical and Dental Equipment Store Built and operate an e-commerce platform covering product management, website administration, SEO, technical content, and customer relations.
Ark Dental / Andisheh Bayesteh Tejarat Co. Tehran 2018 - 2020	Sales Representative, Medical Equipment Consultant, and Content Contributor - Progressed from a trainee role to sales visits, customer consultation, and dental-equipment product support. - Combined technical product knowledge with customer consultation and sales support.
Irandec.com Sep 2018 - Feb 2020	Content Creator (Part-Time) Produced and updated product descriptions, website content, and marketing materials for an online store.

TEACHING, COURSE DEVELOPMENT, AND CREATIVE WORK

Amirkabir University of Technology Nov 2025 6 sessions, 12 hours	Instructor - Practical MATLAB for Neuroscience Designed and delivered an applied course covering MATLAB workflows for neuroscience students and researchers.
Tech Academy academy-tech.ir Dec 2023 - Present	Founder, Instructor, and Website Administrator - Recorded a 40-hour beginner-to-advanced MATLAB course comprising 38 sessions. - Manage website structure, technical operations, educational content, SEO, and course media.
Freelance Istanbul / Remote Aug 2020 - Feb 2023	Multimedia Producer and Content Creator - Produced video, motion-graphics, website, and social-media content for commercial and educational projects. - Managed Nix Broker social channels and produced 15 promotional videos; led a creative team for the Amirkabir International Winter School of MBA.

TECHNICAL SKILLS

Research and experimental Rodent research; stereotaxic workflows; DBS and neuromodulation; behavioral testing; protocol development.	Programming and analysis Python; MATLAB; statistics; repeated-measures modeling; machine learning; signal/image processing; data visualization.
Scientific communication Scientific writing; evidence extraction; publication figures; graphical abstracts; teaching materials.	Medical and digital operations Equipment inspection and troubleshooting; website administration; SEO; e-commerce; video and motion graphics.